

4.1 Microbial Infection

▼ What are the five main microbial agents?

- Viruses
- Bacteria
- Fungi
- Protozoa
- Helminths

▼ What type of organisms are each of these agents?

- Viruses - not cells in their own right
- Bacteria - prokaryotes (no distinct nucleus)
- Fungi - (single-celled) eukaryotes
- Protozoa - (single-celled) eukaryotes
- Helminths - (multicellular) eukaryotes

▼ Why are viruses described as 'obligate parasites?'

- Because they have to infect hosts and use their material in order to reproduce and survive.
- This means they are not found anywhere else in the environment other than in hosts.

▼ What are the two mechanisms used by viruses to divide?

- Cytolysis or budding out of cells

▼ What is the difference between cytolysis and budding?

- Cytolysis is a method of viral particle release after replication which causes the **death of the infected host cell**
- Cytopathy is a method of viral particle release after replication which **does not cause the death of a host cell, but releases particles via budding.**

▼ What advantage does budding have over cytolysis?

- The virus can continue to divide and replicate whilst the host does not recognise it as a pathogen.

▼ What is a retrovirus?

- A **virus that has an RNA genome.**
- This means it requires **reverse transcriptase.**

▼ What is one of the most important steps to take in eradicating an infectious disease?

- Implementing an efficient vaccine programme.

▼ What is special about the genetic makeup of prokaryotes?

- They contain **one copy of each chromosome, instead of two** like us
- This means that a **mutation** to a gene is increasingly likely to have **serious phenotypic effects**.

▼ What is different between the cytoskeleton of prokaryotes and eukaryotes?

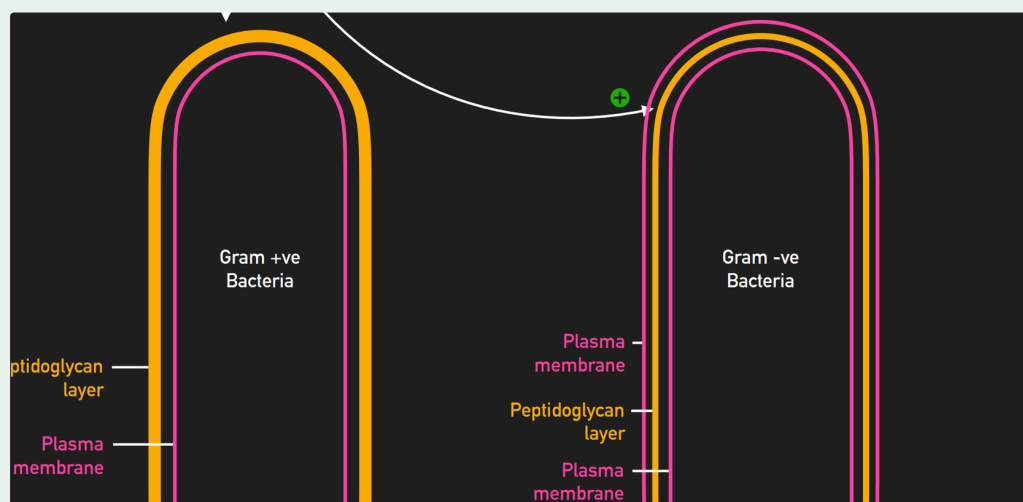
- Prokaryotes have a poor cytoskeleton (most of it comes from the peptidoglycan cell wall)
- They don't have the **actin cytoskeleton**

▼ How do prokaryotes divide?

- By binary fission.

▼ What is the difference between gram-negative and gram-positive bacteria?

- Gram-positive bacteria have **thicker cell walls with more peptidoglycan** than gram negative bacteria
- This can be seen because the **gram-positive** bacterial cell walls **retain the purple crystal violet stain** when it is applied but **gram-negative do not**
- Gram-negative bacteria have **peptidoglycan** which is sandwiched between **two membranes**, but there is only **one of these membranes** in gram-positive bacteria.



▼ What role does the capsule play in bacteria?

- It can allow them to escape phagocytosis by hosts as well as other destructive immune processes as well as protecting them from desiccation (drying up)

▼ What role does the pili play in bacteria?

- It can be used to **attach to certain surfaces**

▼ How can bacteria move other than using their flagella?

- Using the **actin cytoskeleton**

▼ Recall two hospital acquired bacterial infections.

- Clostridium difficile (C. diff)
- Methicillin-resistant Staphylococcus aureus (MRSA)

▼ How do bacterial genomic mutations spread faster than human genomic mutations, despite the fact that both organisms have the same point mutation rates (mutations per number of genes)?

- The rate at which bacteria reproduce is astronomically faster than humans - **they have a shorter generation time**
- A new bacterial generation can take as short as 20 minutes to develop, but in humans, the average generation takes about 26 years to form.
- This means that mutations that occur in bacteria are spread to offspring much more frequently, compared to in humans where the spread can take a number of years.
- Also, the **haploid nature of bacterial cells** means that a mutation spreads far more easily to offspring and can have phenotypic effects more easily, compared to in humans where we have copies of each gene from mum and dad

▼ What is a cutaneous infection?

- One that has its effect through the deep tissue **in the skin**
- For example, it can cause swelling or deep inflammation of skin tissues

▼ What is a visceral infection?

- One that has its effect through the **organs in the body**

▼ How do protozoa divide?

- By **binary fission** or **trophozoite formation**
- Trophozoites are growing stages which result in bursting and the release of protozoa into the environment.

▼ What is a trophozoite?

- A **trophozoite** is the **activated, feeding stage** in the life cycle of certain protozoa such as malaria-causing *Plasmodium falciparum*

▼ What are the two forms that fungi can exist in?

- **Yeasts or filaments.**

▼ Give two examples of viral infections.

- HIV
- Smallpox
- Ebola
- HTLV-1

▼ Give three examples of bacterial infections.

- *Mycobacterium tuberculosis* (TB)
- *Helicobacter pylori* - peptic & gastric ulcers
- Pathogenic *E. Coli* - (urinary tract infections, food poisoning, meningitis etc.)
- *Clostridium difficile*
- Methicillin resistant *staphylococcus aureus*

▼ Give an example of a fungal infection.

- Athlete's foot

▼ Give two examples of protozoa.

- Malaria
- Leishmania

▼ Give three examples of helminths.

- Roundworms
- Flatworms
- Tapeworms
- Hookworms

▼ What characteristic of helminths can make them easier to avoid?

- They are visible to the naked eye.

▼ What does the term 'viral tropism' refer to?

- Viral tropism is the range of hosts that viruses can bind onto

▼ What is a bacillus?

- It is a gram-positive, rod shaped bacteria.

▼ What is a nosocomial infection?

- An infection **acquired in a hospital**.